



Unleash your potential

Module Presenter's Manual *for*

Web Application Development using Python

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Ver. 1.0***

Amendment Record

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1. Introduction

At the end of this course, students will be able to:

- Learn object-oriented Python Web programming
- Working with client-servers, databases, and Web frameworks
- Understand the basics of the Flask framework
- Learn how REST works relative to APIs

2. Information on Session Allocation

Module	Online Hours
Web Application Development using Python	40

Throughout this Presenter's Manual, the module *Web Application Development using Python* will be referred to as **WPYT**.

3. Session Deliverables

To aid the teaching process, following are the deliverables

(Faculties can access eBooks and other components in Onlinevarsity through Aptrack.)

Resources available on Onlinevarsity:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Show Me How - Student can view a step-wise simulation/demonstration of the module related topics.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 WPYT – TL1	Session 2 WPYT – TL2	Session 3 WPYT – TL3	Session 4 WPYT – TL4
2	Session 5 WPYT – TL5	Session 6 WPYT – TL6	Session 7 WPYT – TL7	Session 8 WPYT – TL8
3	Session 9 WPYT – TL9	Session 10 WPYT – TL10	Session 11 WPYT – TL11	Session 12 WPYT – TL12
4	Session 13 WPYT – TL13	Session 14 WPYT – TL14	Session 15 WPYT – TL15	Session 16 WPYT – TL16
5	Session 17 WPYT – TL17	Session 18 WPYT – TL18	Session 19 WPYT – TL19	Session 20 WPYT – TL20

WPYT: Web Application Development using Python

TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	WPYT – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 1 – Overview of OOP Terminology Using Python</u> ➤ Outline the steps to setup Python ➤ Describe the Object-oriented Programming (OOP) concepts ➤ Identify the built-in class attributes ➤ Explain class inheritance ➤ Describe the concept of overriding methods ➤ Describe the concept of overloading methods and operators ➤ Explain multithreaded programming in Python <u>Session 2 – How Does Python Fit into Web Development?</u> ➤ Identify the components of Web application ➤ Describe the advantages of Web development with Python ➤ Explain the procedure of starting a simple server using Python and Flask ➤ Explain Python CGI programming	Web Development using Python-INTL SG – Session 1 & 2 XP – Session 1 & 2 TG – Session 1 & 2
2	WPYT – TL2	All the topics as listed below from Session 3 and Session 4 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 3 – Understand HTML, XML, and JavaScript</u>	Web Development using Python-INTL SG – Session 3 & 4 XP – Session 3 & 4 TG – Session 3 & 4

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain the process involved in Web design ➤ Describe how JavaScript is used in Webpages ➤ Explain the importance of Task queues ➤ Explain Static Site Generator ➤ Explain the steps involved in launching Web2py <u>Session 4 – Python Web Frameworks</u> ➤ Explain Web frameworks and identify different Web frameworks for Python ➤ Outline the differences between Flask and Django ➤ List the steps to install Flask on Windows ➤ List the steps to execute a simple Web application program on Python server	
3	WPYT – TL3	The Try It Yourself Questions from Session 1 to Session 4 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL (Onlinevarsity) Session 1-4
4	WPYT – TL4	All the topics as listed below from Session 5 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 5 – Basic Python Application Structure</u> ➤ Describe Flask Initialization ➤ Explain routes in Flask applications ➤ List the steps to create a Flask application ➤ Explain the Flask's request-response cycle ➤ Describe how to initialize Flask extensions	Web Development using Python-INTL SG – Session 5 XP – Session 5 TG – Session 5
5	WPYT – TL5	All the topics as listed below from Session 6 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL SG – Session 6 XP – Session 6

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 6 – Web Templates</u> ➤ Explain the importance of template engines ➤ Identify the different types of template engines ➤ Describe the popular template engines for Python ➤ Explain the template rendering process in Flask ➤ Identify the extensions for working with Bootstrap and date and time in Flask ➤ Describe the procedure of creating custom error pages in Flask ➤ Describe the functions for obtaining absolute links and static files in Flask	TG - Session 6
6	WPYT – TL6	The Try It Yourself Questions from Session 5 and Session 6 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL (Onlinevarsity) Session 5 & 6
7	WPYT – TL7	All the topics as listed below from Session 7 and Session 8 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 7 – GUI and Web Forms</u> ➤ Describe the significance of Flask Web Forms ➤ Explain Cross-Site Request Forgery (CSRF) protection ➤ Identify form classes ➤ Describe HTML rendering of forms ➤ Explain form handling in view functions ➤ Identify redirects and user sessions ➤ Describe message flashing	Web Development using Python-INTL SG – Session 7 XP – Session 7 TG – Session 7
8	WPYT – TL8	All the topics as listed below from Session 8 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL SG – Session 8 XP – Session 8

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 8 – Database Integration</u> ➤ Define relational databases ➤ Explain how SQL databases store data ➤ Describe NoSQL databases ➤ Explain Python database frameworks ➤ Describe database management using Flask-SQLAlchemy ➤ Explain the different database operations ➤ Explain how to update applications using the Flask-Migrate	TG - Session 8
9	WPYT – TL9	All the topics as listed below from Session 9 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 9 – E-mail Support for Web Applications</u> ➤ Explain the process of sending e-mail and text messages using Python ➤ Describe the process of sending an e-mail using Simple Mail Transfer Protocol (SMTP) and Internet Message Access Protocol (IMAP) servers	Web Development using Python-INTL SG – Session 9 XP – Session 9 TG – Session 9
10	WPYT – TL10	All the topics as listed below from Session 9 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 9 – E-mail Support for Web Applications</u> ➤ Explain e-mail support with Flask-Mail ➤ List the steps to send an e-mail from Web applications	Web Development using Python-INTL SG – Session 9 XP – Session 9 TG – Session 9
11	WPYT – TL11	All the topics as listed below from Session 10 of <i>Web Development</i>	Web Development using Python-INTL

Session No.	Session Title	Session Details	Deliverables' Mapping
		<i>using Python-INTL</i> book should be covered in this session. <u>Session 10 – User Authentication Support for Applications</u> ➤ Identify the different authentication-specific packages and extensions for Flask ➤ Explain password security ➤ Describe how to create the Authentication Blueprints ➤ Explain User Authentication with Flask-Login ➤ Explain the process of registering new users and managing their accounts	SG – Session 10 XP – Session 10 TG - Session 10
12	WPYT – TL12	The Try It Yourself Questions from Session 7 to Session 10 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL (Onlinevarsity) Session 7-10
13	WPYT – TL13	All the topics as listed below from Session 11 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 11 – Application Programming Interface</u> ➤ Explain API concepts ➤ Describe API creation and integration ➤ Define Representational State Transfer (REST) ➤ Explain RESTful services with flask	Web Development using Python-INTL SG – Session 11 XP – Session 11 TG - Session 11
14	WPYT – TL14	The Try It Yourself Questions from Session 11 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL (Onlinevarsity) Session 11
15	WPYT – TL15	All the topics as listed below from Session 12 of <i>Web Development</i>	Web Development using Python-INTL SG – Session 12 XP – Session 12

Session No.	Session Title	Session Details	Deliverables' Mapping
		<i>using Python-INTL</i> book should be covered in this session. <u>Session 12 – WSGI Explorations in Python</u> ➤ Describe Web Server Gateway Interface (WSGI) ➤ Explain WSGI applications ➤ Explain Pylon Web framework ➤ Explain WSGI Servers ➤ Describe WSGI middleware	TG - Session 12
16	WPYT – TL16	The Try It Yourself Questions from Session 12 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL (Onlinevarsity) Session 12
17	WPYT – TL17	All the topics as listed below from Session 13 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 13 – Web Sockets and Socket Programming in Python</u> ➤ Define Web sockets ➤ Describe the significance of Web sockets ➤ Describe the process of implementing Web sockets in Python ➤ Describe the process of creating TCP/IP client and server sockets	Web Development using Python-INTL SG – Session 13 XP – Session 13 TG – Session 13
18	WPYT – TL18	The Try It Yourself Questions from Session 13 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL (Onlinevarsity) Session 13
19	WPYT – TL19	All the topics as listed below from Session 14 and Session 15 of <i>Web Development using Python-INTL</i> book should be covered in this session. <u>Session 14 – Web Application Security</u>	Web Development using Python-INTL SG – Session 14 & 15 XP – Session 14 & 15 TG – Session 14 & 15

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain Cross-site Scripting (XSS) ➤ Describe SQL Injection (SQLi) ➤ Describe Cross-site request forgery ➤ Explain Web Application Security <u>Session 15 – Web Application Testing and Deployment</u> ➤ Describe the importance of unit testing in Python ➤ Identify the unit testing tools in Python ➤ Explain the significance of using flask test client for creating test cases ➤ Explain the procedure of end-to-end testing with Selenium ➤ Identify the deployment options for Web applications ➤ Explain the procedure of using the Heroku platform ➤ Describe the procedure of traditional hosting	
20	WPYT – TL20	The Try It Yourself Questions from Session 14 and Session 15 of <i>Web Development using Python-INTL</i> book should be covered in this session.	Web Development using Python-INTL (Onlinevarsity) Session 14 & 15

6. Library References

➤	Mastering Python by Rick van Hattem
➤	Flask Web Development: Developing Web Applications with Python by Miguel Grinberg
➤	Flask Web Development 2e by Miguel Grinberg

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